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# Innova – M7-PU Modbus quick reference

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## 1 Introduction

This document gives a brief outline of the basic Modbus commands for the following Innova products.

| Internal name | Commercial code(s) | Description              | Applicable to      |
|---------------|--------------------|--------------------------|--------------------|
| M7            | <i>tbd</i>         | Fancoil wall control     | Release 1.5 and up |
| PU            | <i>tbd</i>         | Fancoil on-board control | Release 1.5 and up |

## 2 M7

### 2.1 Quick reference

#### Default configuration

- Serial: 9600-8N1 (PC port)
- Protocol: Modbus RTU
- Modbus Address: 1
- Supported Modbus FCs:
  - FC03: Read holding registers (**max reg. for single request: 3**)
  - FC06: Write single register
- Minimum time between Modbus commands: **150ms**

| Reg. | Bit    | Acronym | Description                                            | Unit   | R/W | Default value | Min | Max |
|------|--------|---------|--------------------------------------------------------|--------|-----|---------------|-----|-----|
| 198  |        | Release | FW release                                             | 0.1    | R   | -             | -   | -   |
| 199  |        | ID      | FW identifier                                          | -      | R   | 1190 (PU)     | -   | -   |
| 550  |        | ADR     | Modbus slave address                                   | -      | R/W | 1             | 1   | 255 |
| 553  |        | PRG     | Program                                                | -      | R/W | -             | -   | -   |
|      | .[2-0] | Mode    | auto=000<br>night=001<br>max=010                       | -      | R/W | auto          | -   | -   |
|      | .3     | Lock    | keys_locked=1<br>keys_unlocked=0                       | -      | R/W | keys_unlocked | -   | -   |
|      | .4     | Stby    | standby=1<br>on=0                                      | -      | R/W | on            | -   | -   |
| 305  |        | SP      | Air temperature set-point                              | 0.1 °C | R/W | 200           | 50  | 400 |
| 312  |        | SP_RH   | Rel.humidity set-point                                 | %      | R/W | 50            | 20  | 90  |
| 556  |        | MAN     | Season selection<br>auto=0<br>man_heat=1<br>man_cool=2 | -      | R/W | auto          | -   | -   |
| 0    |        | T_AIR   | T1 air temperature                                     | 0.1 °C | R   | -             | -   | -   |
| 20   |        | RH      | Relative humidity                                      | %      | R   | -             |     |     |
| 150  |        | Status  | Unit status                                            | -      | R   | -             | -   | -   |
|      | .0     |         | At least one alarm present                             | -      | R   | -             | -   | -   |
|      | .2     |         | Control in stand-by                                    | -      | R   | -             | -   | -   |
|      | .3     |         | Communication error                                    |        |     |               |     |     |

| Reg. | Bit | Acronym | Description                                  | Unit | R/W | Default value | Min | Max |
|------|-----|---------|----------------------------------------------|------|-----|---------------|-----|-----|
| 151  |     | Alarms  | Unit alarms                                  | -    | R   | -             | -   | -   |
|      | .0  |         | Communication error                          | -    | R   | -             | -   | -   |
| 574  |     | TY      | Unit type<br>RP=0, PE=1, PN=2, PS=3,<br>R1=4 | -    | R/W | RP            | -   | -   |

## 2.2 Main commands

| Action |                                                                           | How-to                                                                                   |
|--------|---------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| 100    | Turn unit on                                                              | Write bit <i>Stby</i> ← <i>on</i> to <i>PRG</i> register                                 |
| 101    | Put unit into standby mode                                                | Write bit <i>Stby</i> ← <i>standby</i> to <i>PRG</i> register                            |
| 102    | Select unit fan mode                                                      | Write bits <i>Mode</i> ← <i>auto</i> / <i>max</i> / <i>night</i> to <i>PRG</i> register  |
| 103    | Lock / unlock touchpad                                                    | Write bit <i>Lock</i> ← <i>keys_locked</i> / <i>keys_unlocked</i> to <i>PRG</i> register |
| 104    | Choose room temperature set-point                                         | Write temperature value to <i>SP</i> register                                            |
| 105    | Select season heating or cooling.<br>N.B. Use auto only for 4 pipes unit. | Write register <i>MAN</i> ← <i>man_heat</i> / <i>man_cool</i> / <i>auto</i>              |
|        |                                                                           |                                                                                          |
| 200    | Get room air temperature                                                  | Read register <i>T_AIR</i>                                                               |
| 201    | Get room relative humidity                                                | Read register <i>RH</i>                                                                  |
| 210    | Get room air quality index                                                | Read register <i>IAQ</i>                                                                 |
| 211    | Get TVOC concentration                                                    | Read register <i>TVOC</i>                                                                |
| 212    | Get CO2 concentration                                                     | Read register <i>CO2</i>                                                                 |
|        |                                                                           |                                                                                          |
| 300    | Change Modbus slave address                                               | Write new address to register <i>ADR</i> .<br>The change is immediately effective.       |
| 310    | Check FW version                                                          | Read register <i>Release</i>                                                             |
| 311    | Check FW identifier                                                       | Read register <i>ID</i>                                                                  |

## 3 PU

### 3.1 Quick reference

#### Default configuration

- Serial: 9600-8N1
- Protocol: Modbus RTU (with on-board dip-switch F in position on at power-on)
- Modbus Address: 1
- Supported Modbus FCs:
  - FC03: Read holding registers (**max reg. for single request: 3**)
  - FC06: Write single register
- Minimum time between Modbus commands: **150ms**

| Reg. | Bit    | Acronym   | Description                                            | Unit   | R/W | Default value | Min | Max |
|------|--------|-----------|--------------------------------------------------------|--------|-----|---------------|-----|-----|
| 198  |        | Release   | FW release                                             | 0.1    | R   | -             | -   | -   |
| 199  |        | ID        | FW identifier                                          | -      | R   | 1190 (PU)     | -   | -   |
| 550  |        | ADR       | Modbus slave address                                   | -      | R/W | 1             | 1   | 255 |
| 553  |        | PRG       | Program                                                | -      | R/W | -             | -   | -   |
|      | .[2-0] | Mode      | auto=000<br>night=001<br>max=010                       | -      | R/W | auto          | -   | -   |
|      | .3     | Lock      | keys_locked=1<br>keys_unlocked=0                       | -      | R/W | keys_unlocked | -   | -   |
|      | .4     | Stby      | standby=1<br>on=0                                      | -      | R/W | on            | -   | -   |
| 305  |        | SP        | Air temperature set-point                              | 0.1 °C | R/W | 200           | 50  | 400 |
| 556  |        | MAN       | Season selection<br>auto=0<br>man_heat=1<br>man_cool=2 | -      | R/W | auto          | -   | -   |
| 0    |        | T_AIR     | T1 air temperature                                     | 0.1 °C | R   | -             | -   | -   |
| 1    |        | T_WATER_2 | T2 water temperature                                   | 0.1 °C | R   | -             |     |     |
| 2    |        | T_WATER_4 | T3 water temperature                                   | 0.1 °C | R   | -             |     |     |
|      |        |           |                                                        |        |     |               |     |     |
| 150  |        | Status    | Unit status                                            | -      | R   | -             | -   | -   |
|      | .0     |           | Cooling active                                         | -      | R   | -             | -   | -   |
|      | .1     |           | Heating active                                         | -      | R   | -             | -   | -   |
|      | .2     |           | Water fan stop in cooling                              | -      | R   | -             | -   | -   |

| Reg. | Bit | Acronym | Description                                               | Unit | R/W | Default value | Min | Max |
|------|-----|---------|-----------------------------------------------------------|------|-----|---------------|-----|-----|
|      | .3  |         | Water fan stop in heating                                 | -    | R   | -             | -   | -   |
|      | .4  |         | Fan stop for inadequate T2 temperature                    | -    | R   | -             | -   | -   |
|      | .5  |         | Fan stop for inadequate T3 temperature                    | -    | R   | -             | -   | -   |
|      | .6  |         | Stop for inadequate water temperature trend               | -    | R   | -             | -   | -   |
|      | .7  |         | Antifreeze active                                         | -    | R   | -             | -   | -   |
|      | .8  |         | At least one alarm active                                 | -    | R   | -             | -   | -   |
|      | .10 |         | Standby active                                            | -    | R   | -             | -   | -   |
|      | .11 |         | Master communic. timeout (when using <i>remote mode</i> ) | -    | R   | -             | -   | -   |
|      | .12 |         | Missing H2 probe (only for selected units)                | -    | R   | -             | -   | -   |
|      | .13 |         | Missing H4 probe (only for selected units)                | -    | R   | -             | -   | -   |
|      |     |         |                                                           |      |     |               |     |     |
| 151  |     | Alarms  | Unit alarms                                               | -    | R   | -             | -   | -   |
|      | .0  |         | Communication error                                       | -    | R   | -             | -   | -   |
|      | .1  |         | T1 AIR probe fault                                        | -    | R   | -             | -   | -   |
|      | .2  |         | T3 water probe fault                                      | -    | R   | -             | -   | -   |
|      | .3  |         | Inadequate water temperature                              | -    | R   | -             | -   | -   |
|      | .4  |         | T2 water probe fault                                      | -    | R   | -             | -   | -   |
|      | .5  |         | Inadequate T3 temperature                                 | -    | R   | -             | -   | -   |
|      | .6  |         | Heating resistance high temperature                       | -    | R   | -             | -   | -   |
|      | .7  |         | Fan motor failure                                         | -    | R   | -             | -   | -   |
|      | .8  |         | IN1 switch open                                           | -    | R   | -             | -   | -   |
|      | .9  |         | Inadequate T2 temperature                                 | -    | R   | -             | -   | -   |
|      | .10 |         | Air filter needs cleaning                                 | -    | R   | -             | -   | -   |
|      | .11 |         | Lock for inadequate T2 temperature                        | -    | R   | -             | -   | -   |
|      | .12 |         | Lock for inadequate T3 temperature                        | -    | R   | -             | -   | -   |

## 3.2 Main commands

| Action |                                                                           | How-to                                                                                         |
|--------|---------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| 100    | Turn unit on                                                              | Write bit <i>Stby</i> ← <i>on</i> to <i>PRG</i> register                                       |
| 101    | Put unit into standby mode                                                | Write bit <i>Stby</i> ← <i>standby</i> to <i>PRG</i> register                                  |
| 102    | Select unit fan mode                                                      | Write bits <i>Mode</i> ← <i>auto</i> / <i>max</i> / <i>night</i> to <i>PRG</i> register        |
| 103    | Lock / unlock touchpad                                                    | Write bit <i>Lock</i> ← <i>keys_locked</i> / <i>keys_unlocked</i> to <i>PRG</i> register       |
| 104    | Choose room temperature set-point                                         | Write temperature value to <i>SP</i> register                                                  |
| 105    | Select season heating or cooling.<br>N.B. Use auto only for 4 pipes unit. | Write register <i>MAN</i> ← <i>man_heat</i> / <i>man_cool</i> / <i>auto</i>                    |
|        |                                                                           |                                                                                                |
| 200    | Get room air temperature                                                  | Read register <i>T_AIR</i>                                                                     |
| 220    | Check unit alarms                                                         | Read bits from register <i>Alarms</i> .<br>If the corresponding bit is 1, the alarm is active. |
|        |                                                                           |                                                                                                |
| 300    | Change Modbus slave address                                               | Write new address to register <i>ADR</i> .<br>The change is immediately effective.             |
| 310    | Check FW version                                                          | Read register <i>Release</i>                                                                   |
| 311    | Check FW identifier                                                       | Read register <i>ID</i>                                                                        |

## 4 Remote mode for PU [ext]

|                                                    |          |
|----------------------------------------------------|----------|
| <b>Remote mode for PU [ext] .....</b>              | <b>1</b> |
| What is remote mode?.....                          | 1        |
| How is it enabled?.....                            | 2        |
| Quick Modbus reference.....                        | 2        |
| Recommended control procedure for remote mode..... | 4        |

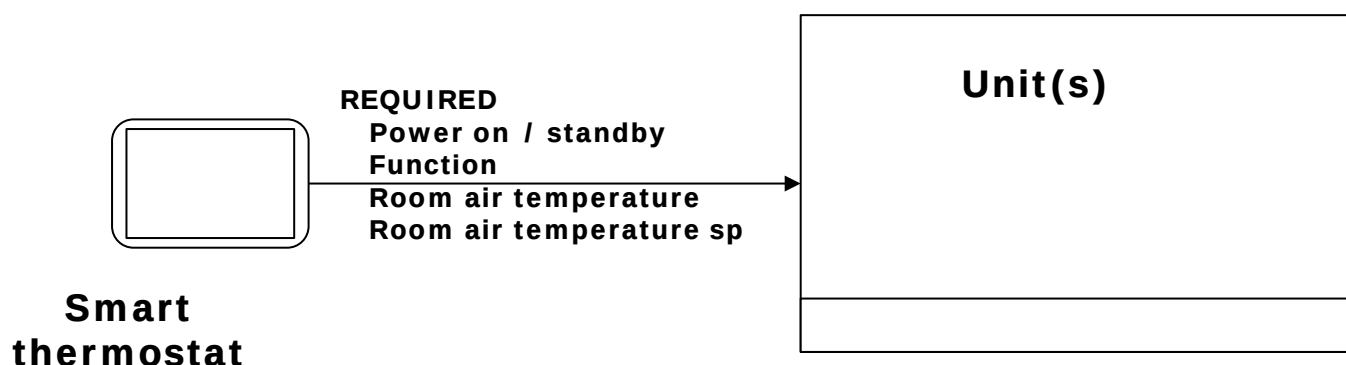
PU series units can be configured for *remote mode* usage.

### 4.1 What is remote mode?

In remote mode, the units are controlled by an external thermostat or supervisor. In this mode, all the information required for environment air regulation is given to the units by the thermostat, that acts as a Modbus RTU/ASCII master.

Each unit configured for remote mode **MUST** receive from the master the following information, at regular intervals:

- room air temperature;
- room air temperature set-point;
- working mode (see [Quick reference](#) for details).



## 4.2 How is it enabled?

Two alternative options:

- A. Write 1 to *REM\_MODE.REM* bit;
- B. Write 1 to *CFG.REM* bit.

### ⚠ Important notes about remote mode activation

- Once the remote mode operation is activated, it can only be deactivated (thus restoring the default unit behavior) by setting the *CFG.REM* bit to 0.
- When a unit is in remote mode, it expects at least one successful communication with the master every 5 minutes, otherwise it will raise an error.

## 4.3 Quick Modbus reference

Default configuration

- Protocol and serial configuration (can be selected by setting the on-board jumper / DIP switch; see manual for details)
  - Option A: Modbus RTU, 9600-8N1
  - Option B: Modbus ASCII, 9600-7E1
- Modbus Address: 1
- Supported Modbus FCs:
  - FC03: Read holding registers (max reg. for single request: 3)
  - FC06: Write single register
- Minimum time between Modbus commands: 150 ms

| Reg. | Bit    | Acronym  | Description                                 | Unit | R / W | Default value | Min | Max |
|------|--------|----------|---------------------------------------------|------|-------|---------------|-----|-----|
| 100  |        | REM_MODE | Remote work mode                            | -    | R/W   | -             | -   | -   |
|      | .[2-0] | PRG      | auto=000<br>night=001<br>max=010<br>min=011 | -    | R/W   | -             | -   | -   |
|      | .3     | -        | -                                           | -    | R/W   | -             | -   | -   |
|      | .4     | REM      | rem_mode_enable=1<br>rem_mode_disable=0     | -    | R/W   | -             | -   | -   |
|      | .[6-5] | -        | -                                           | -    | R/W   | 0             | -   | -   |
|      | .7     | Stby     | standby=1<br>on=0                           | -    | R/W   | -             | -   | -   |

| Reg. | Bit      | Acronym | Description                                    | Unit   | R / W | Default value | Min | Max |
|------|----------|---------|------------------------------------------------|--------|-------|---------------|-----|-----|
|      | .[11-8]  | -       | -                                              | -      | R/W   | 0             | -   | -   |
|      | .12      | CP      | contact_closed=0<br>contact_open=1             | -      | R/W   | 0             | -   | -   |
|      | .[14-13] | Season  | cooling=10<br>heating=01<br>auto_changeover=00 | -      | R/W   | -             | -   | -   |
|      | .15      | TMP     | On board probe=1<br>Remote probe=0             | -      | R/W   | 0             | -   | -   |
| 101  |          | REM_SET | Remote air temperature set-point               | 0.1 °C | R/W   | -             | 50  | 400 |
| 102  |          | REM_TA  | Remote air temperature                         | 0.1 °C | R/W   | -             | -   | -   |
|      |          |         |                                                |        |       |               |     |     |
| 550  |          | ADR     | Modbus slave address                           | -      | R/W   | 1             | 1   | 255 |
|      |          |         |                                                |        |       |               |     |     |
| 552  |          | CFG     |                                                |        | R/W   |               |     |     |
|      | .1       | REM     | rem_mode_enabled=1<br>rem_mode_disabled=0      | -      | R/W   | -             | -   | -   |

## 4.4 Recommended control procedure for remote mode

1. **Activate remote mode.**
2. **At regular intervals\*** - recommended: **30 seconds** (max: 120 s):
  - a. **write REM\_MODE, REM\_SET, REM\_TA** to Modbus broadcast address 0 or to every unit (with unicast addressing)

\*Note: in addition to this, additional commands can be sent asynchronously. For example, on-off commands or set-point changes may be transferred immediately to the units.